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King Khalid International Airport (KKIA) Project

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Project Definition

In order to stay up to date with the Saudi Vision 2030, the project must involve the construction of a new passenger terminal as an expansion of the Riyadh airport King Khalid International Airport (KKIA) while also ensuring that the goals of expanding capacity, increasing operational effectiveness, and improving customer experience is met.

The project scope includes 4 main categories which are project management, design, permitting, and environmental assessments. In addition, the site preparation, infrastructural work, and full-scale construction comes after which covers run way and hanger extensions, apron expansion, underground utility systems, urban landscaping, roads, and parking facilities.

It is also important to mention that the KKIA project includes and adds advanced airport systems such as IT networks, security systems, smart technologies like IoT and AI, passenger service facilities, and baggage handling. The project's terminal design also aims to achieve LEED platinum accreditation, which promises sustainability and low negative environmental effect.

The project has an approximated and expected budget of SAR 850 million which is anticipated to raise annual passenger numbers by about 5 million by the year 2030. In addition, the project will also contain testing, commissioning, and operational transfer.

Project Constraints

The project faces various important limitations that need to be managed diligently to guarantee successful completion

Time Constraints

The KKIA Project has stages that should be achieved to help increase passenger capacity by 2030. However , if any delays were to happen in construction, design approvals, or system integration, this could affect airport operations and the project final completion.

Cost Constraints

We have a budget restriction of about SAR 850 million. So, a strict control of the project costs is needed to reduce costs associated with building, advanced technologies, and environmentally friendly features like the LEED Platinum Certification.

Scope Constraints

The scope of our project includes 4 main elements which are infrastructure, integration of smart systems, extensions for runways and hangers, and the terminal building. If any changes or additions happen to the scope this might cause increased costs and delays, requiring official change management.

Quality Constraints

We are aiming to achieve high standards such as aviation laws safety specifications, and the LEED platinum sustainability requirements for our project. If not achieved, penalties or rework may follow noncompliance of these standards.

Regulatory Constraints

The project must also adhere to the Saudi Arabian building standards and requirements, aviation authorities, and environmental rules, including environmental impact approvals and ICAO criteria.

Resource Constraints

Another important constraint is the availability of skilled labor, construction materials, and advanced new technological systems, especially for specialised areas such as IoT, AI, and airport security infrastructure, which may influence the progress of our project.

Environmental Constraints

The project should follow sustainability goals and eco-friendly building standards. It should also reduce negative environmental impacts such as noise, air pollution, and waste disposal.

Detailed Project Plan

I. Work Breakdown Structure (WBS)

The Work Breakdown Structure for our project, the terminal expansion at King Khaled International Airport in Saudi Arabia, was purposely created to decompose all the required activities and create detailed work packages to achieve a successful outcome in a complex and meticulous project such as an airport extension. This way, all the activities are taken care of and properly organized, ensuring their proper execution, keeping the project control as a priority and a correct schedule development. The WBS follows a hierarchical structure, where each level is more decomposed and more detailed, resulting from a main project objective, multiple work packages with defined deliverables to achieve.

In the project, at the highest level, the first subdivision includes five major components, which are the bases of the terminal expansion. These are the following ones:

- Site Preparation and Infrastructure.
- Urban Planning and Landside Development.
- Construction.
- Services.
- Testing, Commissioning and Handover.

Each of these components is decomposed into low-level work packages until reaching the activities to be done.

The first one, Site Preparation and Infrastructure is the most important one to start the construction and the project itself, as it includes all the work packages and activities related with the preparation of all the required temporary and permanent facilities and purposes needed to start the construction, including land preparation activities, such as ensuring the land's stability, as it needs to be stable and firm to handle the weight of the planes, and the required earthworks before starting the main construction.

The second component is the Urban Planning and Landside Development, focusing mainly on creating and designing an efficient terminal structure, adapting to both the geotechnical and geographical requirements, respecting the land it is being built and also making it comfortable and efficient for the future passengers that may use the terminal. As just explained, its work

packages include:

- Studies and Assessments, making sure that the new terminal does not cause any side effects whatsoever that may impact nature, wildlife and human life quality.
- Parking Structure, ensuring that the passengers have an efficient and reliable way to get to the terminal in a car centered country such as Saudi Arabia.
- Zooning, trying to make the experience for the passengers efficient and reliable.

The next component, once all studies and planning activities are completed and we have ensured that the terminal does not create any harmful effects and meets safety and reliability requirements, is the construction itself. This is the most important part of the project, because it must comply with all the studies and regulations taken into consideration in the previous components. It is divided into MEP and Civil Works, executing all the work and delegating some of the activities into specialised contractors, ensuring high quality standards.

Once the construction is finished and the main components of the terminal expansion are in place, we focus on enhancing the passenger experience and in making the terminal a less hostile environment, that is why the next component is Services. This part of the project includes all the technology services and security services, transforming it into an advanced terminal, using new systems such as artificial intelligence, making the passenger experience easier to handle, more reliable and less stressful with the new security systems, making it both a safe and more convenient place. Another important aspect is the creation of Luxury Services, enhancing the terminal's popularity among the passengers and airlines, and upgrading the airport's prestige.

The final component is the Testing, Commissioning and Handover. This part of the project ensures that all planned activities have been successfully achieved, guaranteeing overall project success. It ensures that all the systems are fully functional and meet all the requirements. This includes diverse testing procedures, ensuring readiness followed by a closing phase before declaring the terminal ready to resume operations.

II. Detailed Discussion of selected Activities required to complete the project

Our project is formed by a great amount of really diverse activities. Out of these, we have decided to analyse 4 main activities that shape our project:

1. Relocate Underground Facilities:

Since the terminal expansion is located in a land area right by the terminal and highways, there are several risks and situations to take into consideration, including the relocation of all underground facilities that may be found during the excavation.

This refers to possible electrical cables, communication cables and water pipelines that supply essential services to the terminal from all these resources. Also, before the construction of the new airport terminal, there must have been some other resources, also including natural ones such as potential archeological sites or undocumented underground features. Because of this, there must be an existing plan taking into account all these possible situations.

If any of these elements happen to be found during the excavations, these should be identified and relocated. It is important that this happens in an early stage, because the construction cannot proceed safely otherwise, compromising the project outcome and causing delays in the project schedule.

2. Noise and Air Quality Impact Study:

Both the construction of a large-scale project such as an airport terminal and the operation of aircrafts can alter the noise and air quality. These are critical factors that we must be aware of and critically assessed. If noise levels and air quality levels rise to a harmful value, the environment can suffer the consequences, and that would be against the guidelines of a successful project as it violates environmental regulations and sustainability requirements.

Because of this, it is mandatory to ensure that the project complies with all the environmental regulations and it does not negatively affect any of the surrounding communities. Quality studies will be regularly done during and after the project to ensure acceptable noise and air quality levels.

3. Install HVAC Systems and Components:

This activity includes the installation of all the heating, ventilation, and air conditioning systems and components in the terminal. This is a critical activity for the building's safety and for the passengers' experience and comfort. The airport is in Saudi Arabia, so taking into consideration the high temperatures that can occur during the summer months, the terminal must have a strong air conditioning system so the passengers can feel comfortable inside. Because of the heat, it is also important to have these systems in all the hallways and corridors connecting the terminal itself with adjacent facilities of the airport such as the parking, bus and train stations among others.

The main challenge to create and implement such an advanced and extensive system is to coordinate it with other systems such as the structural and electrical works. For this reason, several studies and experienced engineers are in charge of developing these systems.

4. Install Smart Airport Devices:

Technology is taking over all modern constructions, however, only high quality and well integrated technology systems stand out and make a difference. This project aims to implement new dynamics and tools that can make this terminal a pioneer in the region. From interactive information displays with face recognition scans that directly guide you to your gate to automated check in counters and passport controls that make the experience easier, faster and more efficient.

It is important that the system is good and does not fail, because one single failure is enough to significantly disrupt the terminal's operations. The cybersecurity threats and system failures are the main risks of this new device's implementation systems. That is why a whole cybersecurity team dedicated to control the correct functioning of this is required, as well as on-site technical staff to solve the problems that may happen and to provide backup solutions, preventing the terminal's operations to be disrupted.

III. Detailed Discussion of Resources required to complete the Project

The completion of the activities in our project requires effective resources. To illustrate it, we are going to focus on the resources required for the four activities previously analysed. Following the 4M's rule, resources are based on: People, Equipment, Materials and Working Capital. Taking into consideration this rule, each activity is going to need a specific combination of these, making each one of them different.

1. Relocate Underground Facilities:

This activity is one of the most technical ones. A group of diverse engineers and technicians is needed to analyse, study and work on uncertain ground conditions. Civil engineers must be the most relevant ones here, but a variety of specialists are required for correct studies and possible issues that may need specialised solutions, such as the

presence of electrical engineers in case there are electrical cables found underground that are connected to the rest of the terminal.

In terms of equipment and materials, excavating machines and ground detection systems may be the most relevant to start the base of the construction and guarantee a safe and stable land to be built. Pipes, cables and covering devices are required for rearranging previously existing structures and cables.

Then, as for working capital, all resources are required to cover mostly equipment usage and labour costs. As an extensive group of technicians is needed and machines to be used are heavy and difficult to transport and move, this activity can be considered one of the most expensive ones. Nonetheless, cost variability is relatively limited, because the work is already well-defined and once the work is properly started in this specific activity, there is not much else that can happen that was not planned.

2. Noise and Air Quality Impact Study:

This activity relies mainly on specialised equipment, as the work is quite methodical. Nevertheless, this equipment must be supported by skilled personnel. So, focusing on manpower, the most important figures that must appear are the environmental engineers and respective data analysts to quantify and interpret the results of the resulting studies. The required machinery includes air quality and noise monitoring systems, which usually are technologically advanced systems with sensors and complex electronics that record the data and information and create warnings whenever the maximum recommended levels are exceeded. To analyse this data, data analysis softwares and computers are also a must.

In this case, material requirements are minimal and financial resources are mainly focused on covering the specialised personnel and technologically advanced equipment.

3. Install HVAC Systems and Components:

This activity is one of the most expensive ones as well due to its required level of precision. It requires a lot of manpower, starting from an extensive group of electrical and electronic engineers, focused on designing plans and schemes to introduce all types of cables and pipes needed for the HVAC Systems to work. Civil engineers and even architects must be needed to ensure the stability of the systems and components to be used.

The materials themselves are also one of the most important resources in this activity, as they are the core of this activity. Air conditioning, heating and ventilation systems

and devices must be installed through the terminal. For these elements to be handled and assembled, machinery is required, including equipment such as lifting machines, cranes and installation tools.

As previously stated, financial resources in this activity are significantly high due to the cost of specialized equipment, engineering teams, required materials and the specific systems and components to be installed in the terminal.

4. Install Smart Airport Devices:

This activity is one of the most technologically advanced ones, so it mainly relies on specialised manpower and technicians such as electronic engineers, computer engineers, data scientists, IT and cybersecurity experts and proper technical staff support as well. Even though at first sight, materials may appear to be the most important resources here, if those materials are not accompanied with a team of specialists, their implementation will not be effective. The material includes devices such as electronic and telecommunication structures and the needed equipment for that are all types of sensors, antennas, servers, facial recognition systems devices and automated units for checking in, automated belts for the luggage among others.

As we can observe, these resources are completely linked to high-end technologies, resulting in high financial costs. The specialised manpower makes it more expensive as well, because really specific engineers and maintenance technicians are needed that are able to manage these new technological items.

IV. Time estimates for the selected Activities

Regarding the time estimates, the same activities we previously chose are going to be analysed. After carefully analysing the uncertainties associated with the activities, realistic time estimations for each one of them have been created.

Activity	Estimated Duration	Explanation
Relocate Underground Facilities	5 to 6 weeks	As underground conditions are unknown, time estimates for this

		activity cannot be determined with high precision. The actions that take the most time can be: excavation, identification, relocation and constant safety checks ensuring a safe and effective execution.
Noise and Air Quality Impact Study	3 to 4 weeks	This activity is one relatively less complex than the rest of the activities. The activity mainly involves installing the measuring devices and having a technical team aware of the situation. Nevertheless, it requires continuous monitoring by them so they can act and alert for possible situations where noise and air quality gets worse. The main things to do is to collect data, monitor it and report it if needed.
Install HVAC Systems and Components	12 to 14 weeks	The installation of HVAC devices is quite complex due to the level of precision they require, both for the materials to be installed and the machinery to be used. The installation of these systems require prior planning and detailed engineering studies where the technicians must ensure a high level of precision. Therefore, there must be a coordination of structural, electrical and architectural works which may make this activity to be one of the longest.
Install Smart Airport Devices	6 to 8 weeks	This activity may appear easier to handle but as the devices and machinery to be used are technologically really advanced, it makes the process to require more study and precision, as these machines are new and they have to be installed, tested and checked for multiple aspects such as the software integration and cybersecurity checks.

Table 1: Time estimates for the selected Activities

V. Cost estimates for the selected Activities

The cost estimates are also related to the previously studied activities and are based on industry benchmarks of the region:

Activity	Estimated Cost	Explanation
Relocate Underground Facilities	SAR 6 million	This activity is one of the most straightforward construction activities to happen in a project, so the work can be executed relatively fast and with common machinery readily available. Its cost is not so high, and the main cost drivers are: excavation equipment, replacements of cables and pipes and labour costs.
Noise and Air Quality Impact Study	SAR 1 million	As this activity is considered relatively less complex, its estimated cost is not going to be as high or expensive as the cost for the rest of the activities. The main costs here are the environmental engineers and data scientists, the quality monitoring devices and the reporting software when threshold limits are exceeded. As these devices are quite common among constructions, there are plenty of options, making it a widely available and easily accessible equipment.
Install HVAC Systems and Components	SAR 80 million	Being the activity with one of the highest estimated time durations also makes it the more expensive activity among the studied ones. Having a big team of engineers, technicians and a specific installation labor behind this activity and having to install the HVAC systems over the whole terminal with a large number of systems and components for them to be effective makes it one of the most expensive activities. Some of the needed elements are: HVAC equipment, ducts, cranes among others.
Install Smart Airport Devices	SAR 40 million	This activity is not difficult to install but its cost is relatively high compared to its installation effort due to using high-end technological devices. These devices are

		advanced and innovative, thus their price is relatively high in comparison to other airport devices. Sensors, servers, automated units for check-in and facial recognition systems combined with IT integration, IoT studies and development and a cybersecurity team results in high overall costs.
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Table 2: Cost estimates for the selected Activities

VI. Gantt Chart

The Gantt Chart for our project is created in the MS file as we can observe.

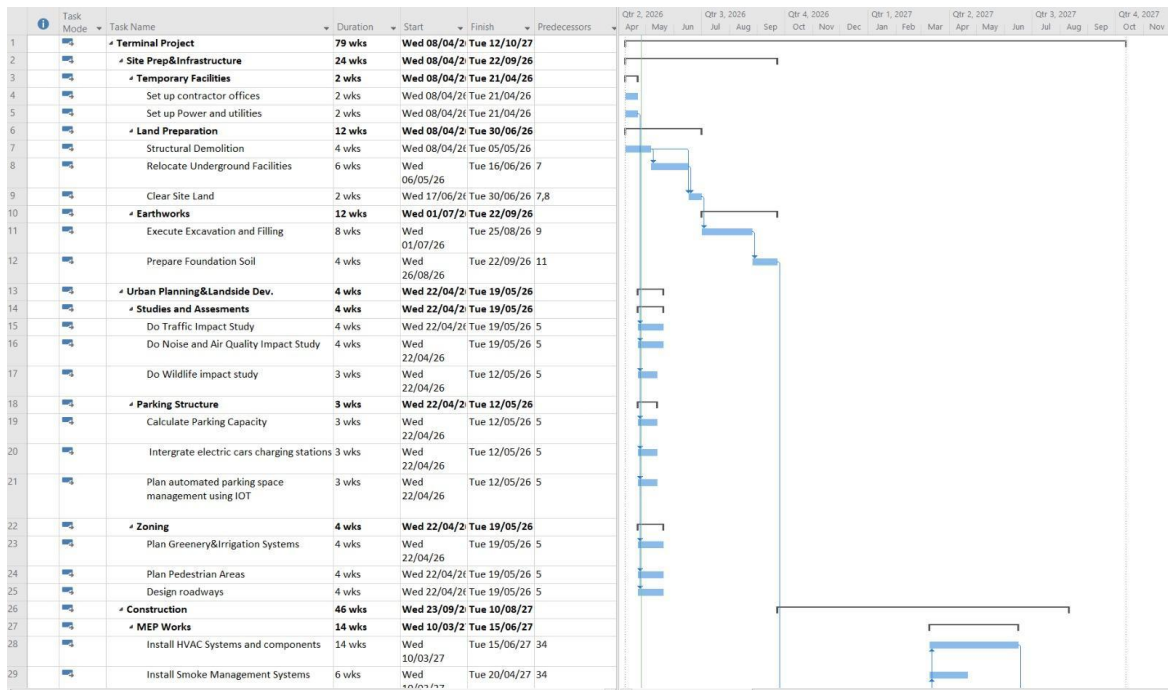


Figure 1. Gantt chart (part I)

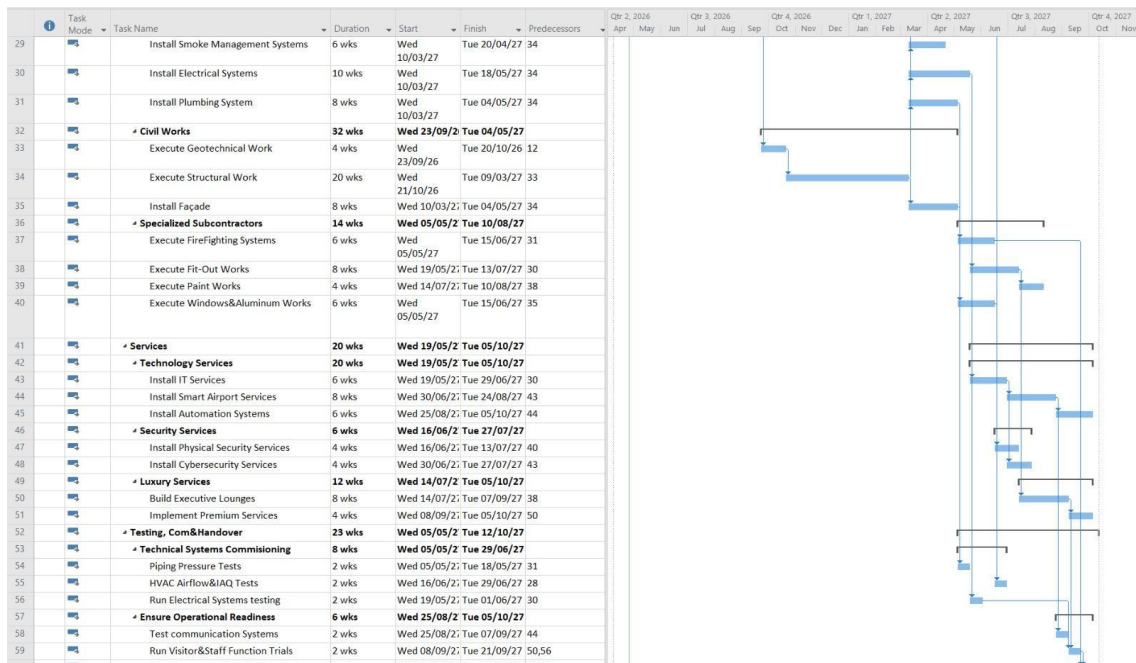


Figure 2. Gantt chart (part II)

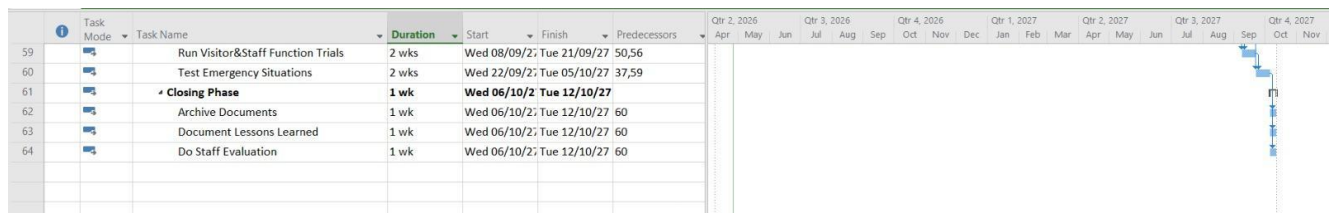


Figure 3. Gantt chart (part III)

We can observe that some activities take significantly longer in comparison to others, so the duration of the activities in the project is quite diverse. Some activities such as executing structural work and installing HVAC systems and components are among the longest due to their complexity, as they require previous studies, a large number of technicians and engineers, a lot of materials and machinery to complete the activities and many more details that lead to an activity to last longer than more simple activities that are more straightforward.

Nonetheless, most of the activities are relatively short, making this project quite dynamic in terms of scheduling. There are some main activities and the remaining ones coexist for shorter periods of time contributing with their objectives, because lasting less does not make the activities less important or irrelevant.

We can also observe all the Early dates from all the project activities. Early Starts and Early Finishes are shown as well as the Gantt graph, where we can start to visualize the critical activities that form the critical path.

VII. Discussion of ES, EF, LS, LF, slack, and project duration for the selected Activities

The detailed discussion of the relevant dates is going to be executed over the sample of activities that we have previously and continuously analysed before:

1. Relocate Underground Facilities:

This activity is one of the first ones to take place and represents a significant stage in the project. Its Early Start is at the beginning of the land preparation phase and its execution depends on the demolition works. Its approximate duration is 6 weeks, so the

Early Finish occurs 6 weeks after the commencement of the activity.

As this activity is part of the critical path of our project, there is no slack, resulting in having a Late Start and Late Finish with the same values and dates as the Early Start and Early Finish. Because of being a critical activity, any delay will directly affect the proposed completion of the project.

2. Noise and Air Quality Impact Study:

This activity is not considered to be part of the critical path, so it has a bit more flexibility in comparison with the previous one. It can be delayed as long as it does not exceed its slack without affecting the overall project completion. As we can observe in the plan made, this activity happens in parallel with other ones. Basically, it has some slack, making its Late Start to occur later.

3. Install HVAC Systems and Components:

The installation of the HVAC Systems and Components is one of the most important activities due to its relevance in the project. Even though it is not part of the critical path, it nearly is. It is dependent on the critical path as it depends on the completion of structural works. Delays in this activity could have a direct impact in the overall project duration, so even though it is not a critical activity, the impacts due to delays in this activity would still affect the project schedule and estimated costs.

4. Install Smart Airport Devices:

This activity takes place in an already advanced stage of the project and although it is important, it is not part of the critical path, meaning that it has some slack that allows some small delays in this activity without disrupting the overall project duration. As its slack is limited, significant delays could affect the testing and commissioning phases of the project, so although it is not a critical one, its duration should be completely controlled.

VIII. Discussion of the Critical Path

The following graph shows the critical path created by all the critical tasks in the project. The mane of each activity, its start and finish dates, percentage of completed work and remaining work time is shown to provide an extensive and correct project schedule.

The critical path consists of the sequence of activities from the project that determine the project duration and have zero slack. If we analyse the activities that form this path, we can see the presence of some really important activities, including some of the longest ones, such as

executing structural works. Some other main activities that we can observe are: relocate underground facilities, which we have been analysing throughout all the project; installing electrical systems; and a lot of final activities such as test emergency situations, archive documents and do staff evaluations. These last activities are part of the last activities happening and usually they form part of the only possible path of completion in the project, and as we can see, they are part of the critical path indeed.

These activities must be carefully controlled and monitored, as they have no slack and any delay would affect the completion date or exceed the estimated costs. That is the main reason why scheduling ahead of time with a realistic plan including all types of resources and elements to complete the activities is needed beforehand. A professional team of engineers is also important to keep track of the completion of the dates and resources used.

CRITICAL TASKS

A task is critical if there is no room in the schedule for it to slip.
[Learn more about managing your project's critical path.](#)

Name	Start	Finish	% Complete	Remaining Work	Resource Names
Structural Demolition	Wed 08/04/26	Tue 05/05/26	0%	0 hrs	
Relocate Underground Facilities	Wed 06/05/26	Tue 16/06/26	0%	0 hrs	
Clear Site Land	Wed 17/06/26	Tue 30/06/26	0%	0 hrs	
Execute Excavation and Filling	Wed 01/07/26	Tue 25/08/26	0%	0 hrs	
Prepare Foundation Soil	Wed 26/08/26	Tue 22/09/26	0%	0 hrs	
Install Electrical Systems	Wed 10/03/27	Tue 18/05/27	0%	0 hrs	
Execute Geotechnical Work	Wed 23/09/26	Tue 20/10/26	0%	0 hrs	
Execute Structural Work	Wed 21/10/26	Tue 09/03/27	0%	0 hrs	
Execute Fit-Out Works	Wed 19/05/27	Tue 13/07/27	0%	0 hrs	
Build Executive Lounges	Wed 14/07/27	Tue 07/09/27	0%	0 hrs	
Run Visitor&Staff Function Trials	Wed 08/09/27	Tue 21/09/27	0%	0 hrs	
Test Emergency Situations	Wed 22/09/27	Tue 05/10/27	0%	0 hrs	
Archive Documents	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	
Document Lessons Learned	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	
Do Staff Evaluation	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	

Figure 4: Critical path and critical tasks

Risk Management

An important aspect we have to consider in regards to our Airport Terminal project is identifying and controlling risks, which are uncertain events that can have a negative effect on our project if they occur. They can create a cost/time overrun, lead to project cancellation, and poor quality which creates customer dissatisfaction. In this section we will discuss possible risk causes and how to address them using the “Risk Management Process”.

- Risk Causes: Below are some risk causes, specific to our terminal project, classified in 2 main categories as below:

1. Internal(Weaknesses):

- ❖ Lack of technical data: If there are missing specifications/plans of the old airport infrastructure, there will be a failure in creating a coherent and consistent extension.
- ❖ Resource/skill shortage: If many resources, especially skilled labor, are not available in the required demand in the country, that issue could slow down the progress and quality of the project. This could happen due to Saudi’s heavy focus and expansion of the mega-project market.
- ❖ Poor Management: For a large and complex project like this, poor management and communication between client, contractor and subcontractors could lead to delays, mistakes in the work, and overall confusion.

2. External(Threats):

- ❖ Environmental factors: Known for its extreme summer conditions, Saudi’s summer heat could lead to work stoppages or damage to materials and equipment. These will cause delays and additional costs incurred.
- ❖ Political and economic fluctuations: If certain events occur in the region, such as political conflicts, there could be direct effects on the project progress. For example, regional conflict could lead to the closure of international trade waterways, causing a delay as well as an increase in resource costs.

- ❖ Stakeholder/customer behaviour: If stakeholder priorities change, there could be a decrease in funding or even pause/cancellation of the project. For example, if the government decides to prioritize and allocate resources to the military defense system instead, the airport project will be put on hold.
- Risk Management Process: Now that we identified the possible causes and effects of risks on our project, we will apply the risk management process to address them. It is a proactive approach that involves following 4 steps to identify, assess, and deal with said risks:
 1. Identifying risks: We first have to identify which risks could have a negative impact on our project. This is done in 3 methods:
 - Brainstorming
 - Risk Breakdown Structure: Breakdown of different categories of where risks can come from; as discussed earlier, can be both internal and external.

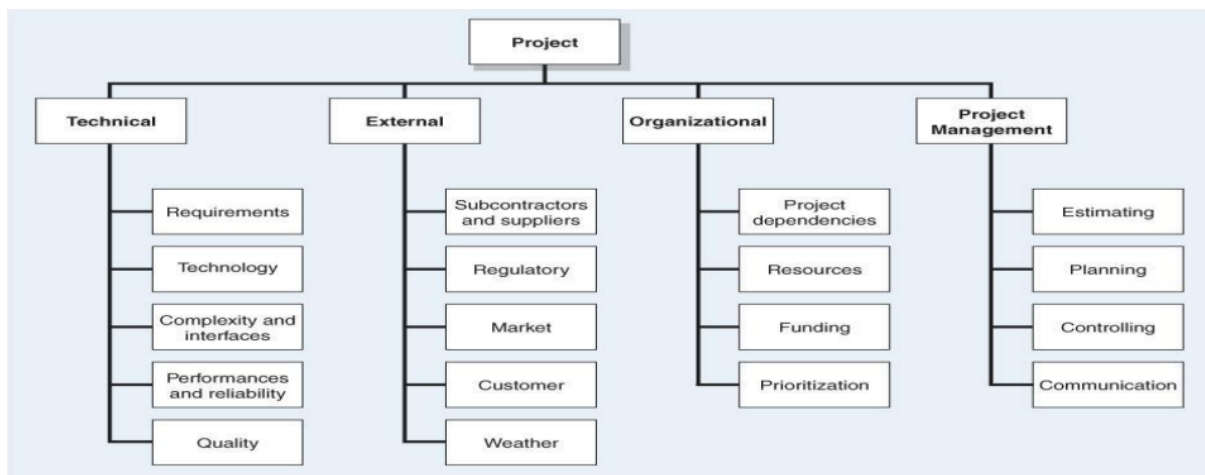


Figure 5: Risk Breakdown Structure (RBS)

- Risk profile: Studying the risks that occurred in past similar projects. For this project, the best risk history is the risks that happened with building the existing airport. All factors will be common such as environment, country codes, regional market, etc.

2. Risk Assessment: This is a crucial step in determining which risks to address first. The rating method is a simple calculation of two factors:

- Risk Severity = Risk Likelihood x Risk Impact Severity
- Risk Assessment Matrix: A table used to list possible risks and describe them according to the following factors and other factors. Below is an example showing a sample of risks relevant to our project:

Risk	Impact	Chance (L,M,H)	Severity (L,M,H)	Action Trigger	Response Plan
Delayed Approvals	Time and budget overrun	M	L	Report from Civil Authority	Rework and resubmitting approval request
Labor Shortage	Time delay and poor quality	M	H	Numerous and persistent unfilled positions	Expedite search for suitable labor in the local and regional market
Extreme weather (sandstorms/heat waves)	Delays and material damages	H	H	Weather forecast	Arrange schedule to account for delays (crash schedule if needed)
Material Inflation	Increased project cost	L	M	Financial market reports/news	Search for cheaper resource alternatives/use contingency funds
Old&new systems integration failure	Synchronization issues between old and new terminals	L	M	Technical issues during and after project handover	Hire technical experts to solve problem quickly

Table 3: Risk Assessment Matrix for KKIA Terminal Expansion Project

3. Risk Response Development: After identifying and assessing the risks in terms of likelihood, severity, and other factors, we will now discuss a plan to reduce the chance of the risk, and its impact if it occurs:

- 4 Main Actions:
 - Avoid: Change plans to completely eliminate exposure to the threat.
 - Mitigate: Plan actions to reduce the chance/impact of the risks.
 - Transfer: Allocate risk to other parties like contractors, insurance companies, etc.
 - Accept/Retain: Set up and use contingency funds to absorb the losses associated with a certain risk.
- Responses to our previous risk examples, with justification:
 1. Delayed Approvals: Mitigate, because you cannot transfer it or accept it (causes delays). Instead, coordinate weekly meetings with authorities to ensure alignment with national codes.
 2. Labor shortages: Mitigate, because you can implement more attractive job offers to attract skilled labor
 3. Extreme weather events: Accept, since it is inevitable in such a region. You can plan to expedite work before known sandstorm periods (form of mitigation) or set up contingencies to cover delays and material damages.
 4. Material Inflation: Transfer, you can transfer to the material suppliers by agreeing on a previous fixed/"frozen" price in advance.
 5. Systems Integration Failure: Mitigate, you can use proven and tried technologies to avoid any problems. In addition, you can plan to test way before handing over to ensure no last-second surprises.

4. Risk Response Control: The final part of the risk management process. It is a constant review of the risk management matrix. It is essential to also identify and assess any new risks, and to implement a risk response plan immediately.

Project Control Methods

“Earned Value Management”

In project management, there are 4 defined phases: Initiating, planning, executing, and closing. In addition, there is a 5th process called ‘Monitoring and Controlling’. It is a continuous process that occurs all throughout the project life cycle. However, we should note that to make sure the project is on track, we have to identify and analyse a few key metrics, within the EVM framework of tracking not just costs&numbers, but the current achieved “value” of the work.

- Key metrics in EVM analysis:
 1. Planned Value: Budgeted cost of the project work at a certain progress point.
 2. Earned value: The budgeted cost of the work that was actually performed.
 3. Actual cost: What was really spent to achieve the work.
- EVM Performance Indicators:
 1. Schedule Performance Index: $SPI = EV/PV$

It indicates the efficiency of which you are progressing in the schedule. If greater than 1, you are ahead of schedule. If less than 1, you are behind on schedule. If equal to 1, you are exactly on track.

2. Cost Performance Index: $CPI = EV/AC$

Measures the efficiency of the project progress in terms of financial terms (cost). If $SPI > 1$, you are under budget, and are getting more value than you invested. If $SPI < 1$, you are over budget, and are getting less value than you put in. If SPI is equal to 1, you are equally efficient.

- Managerial Implications: We have to apply this theory to a few scenarios, in order to understand how to react to changes/challenges and be able to control the project progress to maintain an ideal situation.
 1. If SPI falls below 1: We have to crash critical activities to get back on track with the schedule
 2. If CPI falls below 1: We have to analyze the current financial forecast of the project, and see where materials/resources are increasing in price. In addition, we should analyze project progress to see where materials are not being efficient in their cost-wise

“Return on investment”, such as inefficient machinery or labor.

3. If SPI or CPI are higher than 1: We should take note of why that is the case, and document it so that this lesson can be applied in later projects.
- Additional control methods: We will add a few additional control methods and relate them to the context of our project:
 1. Weekly status and code approval meetings with local Saudi Civil and Regulatory authorities.
 2. In such a large project, implement a “change management system” to ensure changes do not cause gradual scope creep. We cannot afford that in such a complex project.
 3. Use MS Project to track progress and ensure we are not delaying activities on the critical path.

Project Close-out Tasks

In this section, we will highlight the tasks needed to finalize and hand over the airport terminal project. They will be divided into a few major categories:

1. Technical Handover: Making sure the terminal is fully functional before handing it over to the government. In the aviation industry, this is known as “Operational Readiness and Airport Transfer” (ORAT):
 - Snag List: Conducting site walkthroughs and inspections to ensure no minor defects are present.
 - MEP Commissioning: Testing mechanical (HVAC&R), electrical, and plumbing systems to make sure all systems and components are fully functional.
 - Systems Integration Testing: Make sure that the new and old airport logistics and digital systems are fully functional and in sync.
 - Obtain Civil Defense certification from the Saudi Civil Defense, after getting approval that all fire safety systems are clear of any errors and are aligned with

local/national Saudi codes.

- Make sure you avoided scope creep when it comes to technical deliverables. Review your baseline plan.

2. Contractual and Financial closure:

- Final account settlement: Making the final payments to material suppliers and subcontractors.
- Releasing the retained payments/bids to the subcontractors.
- Getting the 'TOC': Taking-Over-Certificate and making sure no issues are detected in the 'Defects Notification Period' (DNP).

3. Administrative/Documentation Closure:

- Archiving project records like project charter, risk profile, specification drawings, etc.
- Handing over user manuals of the terminal's systems to the client.
- Documenting lessons learned (done by the contractor) for future reference

4. Human resources release:

- Coordinating with the company's management to reassign the engineers to other projects in Saudi such as NEOM or the Red Sea Project.
- Performance reviews on all staff involved in the project.
- Clearing the site of labor and resources: Removing any temporary facilities and utilities, returning any borrowed equipment and machinery, and removing any rubble/dirt.

MS Project Reports

PROJECT OVERVIEW

WED 08/04/26 TUE 12/10/27

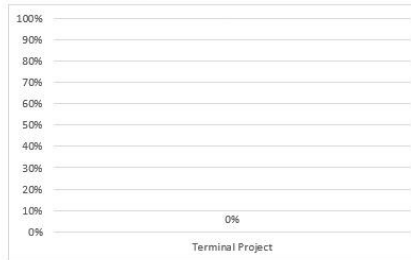


MILESTONES DUE
Milestones that are coming soon.

Name	Finish
------	--------

% COMPLETE

Status for all top-level tasks. To see the status for subtasks, click on the chart and update the outline level in the Field List.



LATE TASKS

Tasks that are past due.

Name	Start	Finish	Duration	% Complete	Resource Names
Set up contractor offices	Wed 08/04/26	Tue 21/04/26	2 wks	0%	
Set up Power and utilities	Wed 08/04/26	Tue 21/04/26	2 wks	0%	
Structural Demolition	Wed 08/04/26	Tue 05/05/26	4 wks	0%	
Do Traffic Impact Study	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Do Noise and Air Quality Impact Study	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Do Wildlife impact study	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Calculate Parking Capacity	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Intergrate electric cars charging stations	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Plan automated parking space management using IOT	Wed 22/04/26	Tue 12/05/26	3 wks	0%	
Plan Greenery&Irrigation Systems	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Plan Pedestrian Areas	Wed 22/04/26	Tue 19/05/26	4 wks	0%	
Design roadways	Wed 22/04/26	Tue 19/05/26	4 wks	0%	

Figure 6: Project Overview

This report provides a high level summary of the King Khalid International Airport terminal extension schedule. It outlines the complete project lifecycle, from the initial date in April 2026, all the way to the operational handover estimated in late 2027. The dashboard serves as the baseline for tracking the overall progress, through different phases to completion, with high level schedule alignment.

CRITICAL TASKS

A task is critical if there is no room in the schedule for it to slip.

[Learn more about managing your project's critical path.](#)

■ Status: Late ■ Status: Future Task

Name	Start	Finish	% Complete	Remaining Work	Resource Names
Structural Demolition	Wed 08/04/26	Tue 05/05/26	0%	0 hrs	
Relocate Underground Facilities	Wed 06/05/26	Tue 16/06/26	0%	0 hrs	
Clear Site Land	Wed 17/06/26	Tue 30/06/26	0%	0 hrs	
Execute Excavation and Filling	Wed 01/07/26	Tue 25/08/26	0%	0 hrs	
Prepare Foundation Soil	Wed 26/08/26	Tue 22/09/26	0%	0 hrs	
Install Electrical Systems	Wed 10/03/27	Tue 18/05/27	0%	0 hrs	
Execute Geotechnical Work	Wed 23/09/26	Tue 20/10/26	0%	0 hrs	
Execute Structural Work	Wed 21/10/26	Tue 09/03/27	0%	0 hrs	
Execute Fit-Out Works	Wed 19/05/27	Tue 13/07/27	0%	0 hrs	
Build Executive Lounges	Wed 14/07/27	Tue 07/09/27	0%	0 hrs	
Run Visitor&Staff Function Trials	Wed 08/09/27	Tue 21/09/27	0%	0 hrs	
Test Emergency Situations	Wed 22/09/27	Tue 05/10/27	0%	0 hrs	
Archive Documents	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	
Document Lessons Learned	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	
Do Staff Evaluation	Wed 06/10/27	Tue 12/10/27	0%	0 hrs	

Figure 7: Critical Path

As shown in Figure 7, the critical path for the King Khalid International Airport terminal extension follows a strict sequential logic from the site preparation phase to the final handover. All these activities dictate the absolute minimum duration of the project, any delay in these specific sequential tasks, especially the final stages which rely on completing the Executive Lounges to allow for Visitor & Staff Function Trails, because these specific tasks possess zero slack, and will directly delay the final passenger terminal opening and violate the project's time constraints.

The milestone dashboard and task burndown trajectory for the project baseline, is shown in Figure 8. This line graph demonstrates the planned reduction of remaining project tasks, indicating a steady construction planned to start in April 2026 and concluding in October 2027. Because this schedule represents the pre-execution baseline, the late and completed milestone categories reflect no recorded progress yet. This establishes a zero baseline schedule which management teams can use to track progress and performance and identify variances in schedule once the construction begins.

The milestone dashboard and task burndown trajectory for the project baseline, is shown in Figure 8. This line graph demonstrates the planned reduction of remaining project tasks, indicating a steady construction planned to start in April 2026 and concluding in October 2027. Because this schedule represents the pre-execution baseline, the late and completed milestone categories reflect no recorded progress yet. This establishes a zero baseline schedule which management teams can use to track progress and performance and identify variances in schedule once the construction begins.

Conclusions

In conclusion, undergoing the execution of an advanced and highly costly project such as our King Khalid Airport Terminal extension, requires attention to detail due to its sheer complexity. Project managers have to juggle a large deliverables breakdown, countless risks on the project objective, and a successful execution of the project objectives without scope creep, time delays, or budget/cost overruns.

Lessons Learned from our side as students:

1. Identifying all parties involved in construction projects, and getting familiar with the nature and know-how of the interactions between them, such as contractual and financial implications.
 2. Understanding how regional codes and physical environments can affect a project in unique ways.
 3. Applying the fundamentals of risk management, which is a highly critical element of viewing a project.
 4. Being able to create a detailed work breakdown structure (WBS) of a major project's deliverables from a contractor's point of view. In addition, we learned to accurately list and estimate the cost and durations of work packages and activities for the WBS we created.
-
- Next steps: Researching different projects and getting acquainted with how our elements of analysis we used for this project will apply to other ones. In addition, we

will undergo the examination for the CAPM certification, to be able to officially “prove” our newfound project management skills.

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